Table 4—Heterogeneity in Treatment Effects by Presence of Suboptimal Baseline Preferences

| | Four-year college (1) | Adjusted graduation rate (2) | BA degree (5 years) (3) |
|---|---|---|---|
| Treatment × suboptimal index | 0.0599 | 1.547 | 0.0438 |
| | (0.0222) | (1.123) | (0.0246) |
| Treatment × inverse-covariance weighted index | 0.0352 | 0.9540 | 0.0298 |
| | (0.0205) | (0.8440) | (0.0174) |

*Notes:* Table provides treatment effect estimates on various outcomes as well as how those treatment effects vary with the intensity of suboptimal baseline college preferences. Each column contains estimates from a separate regression of a dependent variable (in columns) on a treatment indicator variable and treatment indicator variable interacted with an index of the intensity of suboptimal baseline college preferences, controlling for site-by-cohort (i.e., risk set) indicators as well as the covariates indicated in Table 1. The intensity of suboptimal baseline college preferences is captured by an index that combines information from 12 different proxies for suboptimal baseline preferences. The proxies include listing a school with a below-median six-year graduation rate (53.8 percent) as well as the share of institutions listed with low graduation rates, listing a school with a below-median six-year graduation rate and a high net price (over $10,000) as well as the share of institutions listed with low graduation rates and high net prices, listing a school for which the residual from a regression of graduation rate on individual GPA and imputed SAT falls in the bottom 25 percent as well as the share listed meeting those conditions and the opposite of the average residual across baseline schools listed, listing a school in Barron's categories 4–6 (i.e., "Less Competitive," "Noncompetitive," "Special"), listing a for-profit institution, listing a school with median earnings for individuals from low-income families ($0–$30,000) below the bottom quartile in Massachusetts or New York as well as the share listed meeting those conditions, and not listing a single "match" institution (i.e., an institution for which one's predicted SAT fell between the twenty-fifth and seventy-fifth percentiles). The summary index was generated by first standardizing each measure to have mean zero and standard deviation one, adding the measures together, and then restandardizing. Robust standard errors in parentheses.

correlation of measures. If some of the measures are highly correlated, it is *possible* that the equally weighted index is in some sense double-counting the same information (and underweighting other information). To account for this potential double-counting, we consider a second index that weights each standardized measure by the inverse covariance matrix (Anderson 2008). We demonstrate robustness to other aggregation approaches in Supplemental Appendix D.[23]

In Table 4, we show that the effects of advising on measures of degree receipt are significantly larger for those who appear to be less informed at baseline. For instance, a 1 standard deviation increase in suboptimal baseline preferences corresponds to a 3 to 4 percentage point larger effect of intensive advising on degree attainment, roughly half the magnitude of the average treatment effect. The larger increases in degree receipt are preceded by similarly sized increases in the quality of initial enrollment (i.e., four-year enrollment and adjusted graduation rate), supporting the notion that the effects on bachelor's degree attainment are causally mediated by improvements in school choice.[24] The interaction effect estimates using the

[23] In the Supplemental Appendix, we describe the other approaches we explored (e.g., the use of principal component analysis and factor analysis to generate indices), discuss trade-offs between these approaches, and present results from using each index to explore treatment heterogeneity by baseline preferences.

[24] The interaction effects on four-year enrollment are significant across index measures. While the coefficients for the adjusted graduation rate are not significant at conventional levels, the positive direction of the coefficient further suggests that improvements in school choice are a key mechanism. We show that our conclusions are largely robust to the measures included in both indices. Supplemental Appendix Figures D1 and D2 contain the distribution of interaction effect estimates generated when dropping any three measures. For the four-year-enrollment and bachelor's-degree-within-five-years outcomes, the interaction effect estimates are fairly tightly clustered around the effects reported in Table 4; this is also true for the adjusted graduation rate outcome with the equally weighted index. The interaction effect estimates for the adjusted graduation rate with the inverse-covariance weighted index are somewhat more dispersed, with a minority of estimates suggesting very small (less than 0.5 percentage points) negative interaction effects.